

STRIDE Battery Status Monitoring Board

Test Plan

Project	STRIDE	Board / Assembly	Battery Status Monitoring Board
Core Device	LTC2944	Prepared By	Maurice Otieno
Date		Reviewed By	
Board Serial / ID		Test Station	ECE 519

1. Purpose

This document defines the bench bring-up procedure and Acceptance Test Procedure for the STRIDE Battery Status Monitoring Board before subsystem integration. The intent is to verify that the board is safe to power, that its input and regulator stages behave correctly, and that the LTC2944 is reachable over I²C, and that voltage, current, and charge measurements behave correctly under representative operating conditions.

A separate test plan is provided for a dynamic load test that uses motors and Battery rather than bench supply. This is used for the initial bench bring up.

2. Scope

This procedure applies to the STRIDE battery monitoring PCB built around the LTC2944 high-side battery gas gauge architecture. It covers the following phases of board validation:

- Unpowered continuity and sanity checks
- First power-up from a bench source
- Regulator and local rail verification
- I²C communication verification
- Static bus-voltage verification
- Shunt-path and current-sense verification
- Coulomb-count and charge-accumulation verification
- Short board-level soak test

3. Reference Requirements

The battery monitoring board design intent used in this procedure assumes a 24 V battery system, up to 30 A maximum design current, LTC2944-based high-side sensing, a 1 m Ω shunt resistor, and support circuitry including a local regulated logic rail, I²C pull-ups, and decoupling capacitors.

4. Board Under Test

4.1 Core Device

LTC2944 battery gas gauge

4.2 Intended Functional Role

The board measures and reports battery status by monitoring battery bus voltage, shunt voltage across a precision current-sense resistor, inferred current and accumulated charge implemented by the host controller.

4.3 System-Level Design Intent

The board is intended for a 24 V STRIDE battery system with current demand up to no more than 30 A maximum design current, using a 1 m Ω shunt resistor in a high-side current-sensing arrangement.

4.4 Principal Functional Blocks

- Battery positive input – this is the sense path
- Battery negative – this is the return reference.
- Precision shunt resistor
- LTC2944 sensing and I²C interface
- Local regulated rail for logic-side support circuitry
- I²C pull-ups
- Status LEDs as indicators

5. Required Equipment

- Bench DC power supply (The supplies can only deliver 120 W) of power
- Digital multimeter
- Oscilloscope
- 12 ohm 10W power resistor (An electronic load could work)
- Logic host or microcontroller capable of I²C communication (STM32H7A3ZI-Q)
- Insulated test terminals and probes
- Safety glasses

6. Safety Prerequisites

- Confirm that a bench supply is used for first energization.
- Wear safety glasses.
- Keep a power disconnect method within arm's reach.
- Do not rewire the board while energized.
- Do not connect the board directly to the full battery pack until bench tests have passed.
- During current-path tests, ensure all wiring and fixtures are rated for the expected current.
- Treat all exposed conductors as energized once power is applied.

7. General Pass / Fail Definitions

Pass: Measured values are within stated limits, no unsafe condition occurs, communication and measurement behavior are consistent and repeatable, no abnormal heating, smell, smoke, or visible damage occurs.

Fail: Any stated limit is violated, a short or unsafe event occurs, communication is missing or unstable, or measurements are clearly incorrect or not repeatable.

8. Bring-Up Procedure

8.1 Unpowered Electrical Checks

Test definition / method: With all power disconnected, verify that there is no hard short on the main input, that the regulator output is not shorted, and that the I²C nets are not shorted to each other or to the wrong rail.

Procedure:

- Verify continuity of the shunt current path from input side (BAT+) to load side. Do this by measuring the resistance from battery side lead to load side lead of the resistor.
 - Use the Digital multimeter (DMM) to confirm shunt resistor value by performing a continuity test between the supply terminals and the sense terminals.
 - Turn the knob on the Digital multimeter to select the continuity test. You should hear a beeping sound for any resistance value less than 50 Ohms. This is a vague value test to confirm that we have a very low resistance component (1 milli-ohms).
- Confirm there is no hard short between input power and ground.

- With DMM still set to perform continuity, probe the battery side lead with the DMM while holding the other probe to GND. You should not hear any beeping sound this time. If you hear a beeping sound, we have a short.
- Measure resistance from regulated rail to ground.
- To measure the resistance, turn the knob on DMM to select resistance. Set the measurement to take DC readings. By default, the DMM is set to read AC, ensuring you switch to DC by pressing the select button on the upper left area of the DMM. You can also use a continuity test here.
- Measure resistance from SDA to ground and SDA to regulated rail.
- Measure resistance from SCL to ground and SCL to regulated rail.
- Measure resistance between SDA and SCL.
- Verify that the LTC2944 sense pins are connected to the intended sides of the shunt and not swapped.
- Perform one last continuity test between the sense pins of the LTC2944. The pins should be in a specific orientation and you should not hear a beeping sound.

Pass criteria: No hard short exists across the main input, the regulated rail is not shorted, SDA and SCL are not shorted together, and the shunt path matches design intent.

Fail criteria: Any hard short exists, I²C lines are shorted together or to the wrong rail, or the shunt/sense path is open, swapped, or miswired.

8.2 First Live Power-Up

Test definition / method: Energize the board from a current-limited bench source with no heavy external load connected.

Procedure:

- Power up the board by connecting the 24 v pin to the positive supply of the bench.
 - Connect the GND pin on the board to the negative supply.
- Set the bench supply to a safe initial voltage appropriate for the board input stage. Begin with a voltage above just above 5 volts. The supply on the bench at AEC 519 will pull a suitable amount of current to power the board. You should however use the board current setting knob to limit the current for this first energization to no more than 0.3 A. The battery status monitoring board does not consume that much anyway.
- Observe the ON LED. It should light bright red at this point.
- Turn off bench supply.
- Connect the Supply terminals of the shunt resistor to the same bench power supply. Connect the load side terminal to a high power (10W) resistor to the breadboard reference ground. Connect the Battery side resistor terminal to the supply positive power supply terminal.
- Turn on the power supply. Set the voltage to 5 v and observe input current.
- If startup is clean, increase input voltage to nominal operating level i.e. 24 v.

Verify that no component exhibits abnormal heating, odor, smoke, or distress.

Pass criteria: The board powers without sustained current limiting, and no smoke, smell, or abnormal heating occurs. ON LED lights bright RED

Fail criteria: Persistent current limiting, unstable startup, visible distress, or unsafe behavior occurs.

Inspection result: ☐ Pass ☐ Fail

8.4 Powered Rail Sanity Check

Test definition / method: With the board powered, verify the input rail and local regulated rail.

Procedure:

- Measure the local regulated rail to ground. The board is supplied with test pads for the power rail and the regulated 3.3v rail. Probe this with the DMM. Ensure the DMM reference ground is connected to the same circuit ground.
- Verify that the LTC2944 supply pin sees the expected operating voltage. Probe the supply pins with DMM to ensure voltages close to the 24-v nominal.
- Verify that the I²C pull-up rail matches the expected 3.3 v logic voltage.

Pass criteria: Input supply is at the intended value, the local regulated rail is stable 3.3v , and the logic-side rail matches design expectation 3.3v .

Fail criteria: Missing, incorrect, unstable, or noisy supply rails are observed.

8.5 I²C Communication Verification

Test definition / method: Attach the intended logic host and verify that the LTC2944 is reachable over I²C and responds reliably.

Procedure:

- Connect NUCLEO- H7A3ZI-Q (Based on the STM32H7A3ZI-Q pins PB6 and PB7 or any other host) to SDA, SCL, and GND. Connect one of the MCU ground pins to the circuit reference ground.
- Power the board at 24v input voltage.
- Run an I²C scan and record the discovered device address.
- If you choose to use the host device mentioned above, run the Battery read program found at https://github.com/ECE492-Lafayette-College-2026-STRIDE/STRIDE_MCU/tree/main/Core/Src/App/Battery_Monitor. If not, use a program compatible with your host for the LTC2944.
- Monitor the readings using a UART connection via any of the boards UART peripherals and the serial monitor.
- Verify that communication remains stable over repeated transactions on the serial monitor.

Pass criteria: The LTC2944 is detected at the intended address, register reads are valid and repeatable, and any read-back values are correct.

Fail criteria: Device is not detected, communication is intermittent, or register values are invalid or inconsistent.

8.6 Static Bus-Voltage Measurement Check

Test definition / method: Verify that the board reports a sensible battery or input voltage with no significant load current flowing.

Procedure:

- Apply 24 v to the board.
- Measure the actual input voltage with a calibrated DMM.
- Read the LTC2944 bus-voltage register by running the program earlier mentioned.
- Record the measured value) and the reported value (this should appear on the serial monitor).
- Repeat at a minimum of three input voltages spanning the expected operating range.

Pass criteria: Reported bus voltage tracks the applied voltage, agrees with the DMM within the defined tolerance, and changes with applied voltage.

Fail criteria: Reported voltage is missing, unstable, grossly inaccurate, inverted in trend, or clipped unexpectedly.

8.7 Shunt and Current-Sense Verification

Test definition / method: Pass a known current through the current path and verify that the sensed shunt voltage and inferred current behave correctly.

Procedure:

- With the board powered to 24v, measure the voltage across the 10 W power resistor. Ignore the shunt resistor, a very minimal voltage drop occurs across it.
- Calculate the current across the Power resistor using ohms law.
- Read the LTC2944 current data from the program you ran on the MCU.
- Compare reported current against the expected value based on the 1 mΩ shunt and measured load current.
- Reduce supply voltage to 23 v and then turn it up a notch to 29v. Repeat procedure and above.

Pass criteria: Current reading scales correctly with applied load current, and direction, sign, and magnitude are consistent with the test condition.

Fail criteria: Current reading is missing, grossly inaccurate, wrong-sign, unstable, or non-repeatable.

8.8 Coulomb Count / Charge Accumulation Verification

Test definition / method: Verify that the board accumulates charge consistently over time during a controlled current-flow interval.

Procedure.

- Reset the charge accumulator as appropriate by rerunning the program.
- Apply 24 v across the power resistor. Set a timer on your phone but don't start it yet.
- Read the Accumulated charge register (ACR) and immediately start your timer.
- Maintain a timer for 60 seconds and then read the ACR again.
- Compute the expected charge transfer from current multiplied by time. Compute accumulated charge by subtracting initial from final value of the accumulated charge register.
- Repeat for at least one additional current or time interval.

Pass criteria: Accumulated charge changes in the correct direction and is consistent with the applied current-time product within defined tolerance.

Fail criteria: Charge accumulator does not change, changes in the wrong direction, or disagrees significantly with the expected result.

8.9 Short Soak Test

Test definition / method: Operate the board for 10 to 15 minutes at 24v power conditions while periodically polling measurements.

Procedure:

Execute procedures 8.6, 8.7 and 8.8 for 15 minutes

Pass criteria: The board operates continuously without reset, unsafe heating, or communication failure, and measurements remain stable throughout the run.

Fail criteria: Any reset, lockup, overheating, or intermittent behavior occurs.

8.9 Over voltage and Overcurrent Sense

Test definition / method: Operate the board for 10 to 15 minutes at 24v power conditions while periodically polling measurements but now setting the current to 1 A and Voltage to 12 V

Procedure:

Execute procedures 8.6, 8.7 and 8.8 for 15 minutes

9. Bench ATP Table

Test ID	Test Name	Test Definition / Method	Pass Criteria	Fail Criteria
BM-01	Logic rail short check	Check regulated rail and I ² C nets for shorts.	No rail or bus short exists.	Regulated rail or bus short present.
BM-03	First live power	Power board from current-limited DC source.	Safe startup, no distress, no sustained current-limit event. Power LED On	On LED off, smoke, smell, or unsafe behavior.
BM-04	Rail verification	Verify input rail at 24 and regulated rail at 3.3 v under power.	Rails are present and within 3.3v range.	Incorrect, or unstable rail.
BM-05	I ² C detection	Scan and communicate with LTC2944.	Device detected with correct address and communication repeatable.	Device not detected or unstable communication.
BM-06	Bus voltage measurement	Compare reported bus voltage with DMM.	Reported voltage matches expected trend.	Missing, grossly inaccurate, or unstable reading which does not match voltage trend .
BM-07	Low-current sense test	Apply 0.4 A, verify reading.	Current reading matches expected reading and does so repeatedly.	Wrong magnitude, or unstable.
BM-8	Multi-point current test	Repeat current verification at several current levels.	Readings track applied current across range.	Nonlinear, clipped, or erroneous behavior.
BM-9	Coulomb count test	Apply known current for known time and compare charge accumulation.	Charge count changes correctly and within tolerance.	No accumulation, wrong direction, or poor agreement.
BM-10	Short soak test	Run 10 to 15 minutes with periodic polling.	Continuous stable operation.	Reset, lockup, overheating, or fault.

10. Data Recording Sheet

Test ID	Notes on expected behaviour	Result	Pass/fail
BM-01	DMM beeps on shorts	No shorts	Pass
BM-02	Input - 24 Regulated rail - 3.3	23.9 3.28	Pass
BM-03	ON LED - Bright red	ON LED - Bright red	Pass
BM-04	Serial monitor prints out some voltage and current	Serial monitor output	Pass
BM-05	5v 24 v	5.01 23.92	Pass
BM-06	0.4 A	0.423	Pass
BM-07	0.4 A	0.423	Pass
BM-08	Current scales linearly with voltage	Current scales linearly with voltage	Pass
BM-09	0.1		Pass

11. Final Disposition

Overall board status: Pass

Required rework / notes before integration:

Tester signature: _____ Date: _____

Witness / reviewer: _____ Date: _____